SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Case Study

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Annexure A – Answer Key

Short Answer Test: CO3 – Bayesian Learning & Probabilistic Models

Q1. Probabilistic Diagnosis Model for a Hospital System

Conceptual Understanding

Bayes' Theorem provides the mathematical foundation for updating the probability of a disease given observed symptoms: P(Disease|Symptoms) = [P(Symptoms|Disease) · P(Disease)] / P(Symptoms). Here P(Disease) is the prior probability derived from population health statistics, P(Symptoms|Disease) is the likelihood learned from historical patient records, and P(Disease|Symptoms) is the posterior used for diagnosis. A Naive Bayes classifier extends this by assuming conditional independence between symptoms given the disease, which simplifies computation to a product of individual symptom likelihoods, while a Bayesian Belief Network (BBN) relaxes this assumption by explicitly encoding a directed acyclic graph of conditional dependencies among symptoms, risk factors, and diseases.

Accuracy of Answer

For diagnosis, a BBN node structure would connect root nodes (age, genetic predisposition, lifestyle) to intermediate nodes (underlying conditions) to leaf nodes (observable symptoms such as fever, cough, fatigue). Each node stores a Conditional Probability Table (CPT). Inference proceeds by entering observed symptom evidence and propagating it through the network using algorithms such as variable elimination or the Gibbs sampling approach, producing a posterior distribution over candidate diseases rather than a single deterministic label.

Application of Knowledge

- Naive Bayes: fast, works well with limited data and many symptoms, but its independence assumption can misestimate probabilities when symptoms are correlated (e.g., fever and chills often co-occur).
- Bayesian Belief Network: captures true conditional dependencies (e.g., cough depends on both smoking history and infection status), giving more clinically realistic and explainable predictions at the cost of higher computational and data requirements for structure/parameter learning.
- Probability learning (updating priors/likelihoods as new patient data arrives) allows the system to adapt to population drift, seasonal disease patterns, and rare-disease under-representation.

Completeness of Response

Uncertainty is handled explicitly because the output is a probability distribution over diseases rather than a hard classification, allowing clinicians to see confidence levels and request further tests when posterior probabilities are close. Missing symptom values are handled naturally in a BBN by marginalizing over the unobserved variable instead of requiring imputation.

Clarity & Presentation

In summary, a hybrid approach is recommended: Naive Bayes as a fast first-pass screening layer, escalating ambiguous or high-risk cases to a full Bayesian Belief Network for dependency-aware diagnosis. Real-world constraints such as limited labelled clinical data, class imbalance (rare diseases), interpretability requirements for doctors, and the cost of false negatives in a medical setting must all guide model selection and threshold tuning.

Q2. Spam Detection System using Naive Bayes and Bayes Optimal Classifier

Conceptual Understanding

Emails are represented as feature vectors (e.g., bag-of-words or TF-IDF over vocabulary terms). Naive Bayes computes $P(Spam|Words) \propto P(Spam) \cdot \prod P(word\_i|Spam)$, assuming word occurrences are conditionally independent given the class. The Bayes Optimal Classifier instead combines predictions from every hypothesis in the hypothesis space, weighted by its posterior probability, to output the class with the maximum weighted vote — theoretically the best possible classifier for a given hypothesis space, though computationally intractable for large spaces, which is why Naive Bayes is used as a practical approximation.

Accuracy of Answer

Maximum Likelihood Estimation (MLE) selects word-probability parameters that maximize the likelihood of the observed training emails, which can overfit rare words that appear only in a few spam emails (zero-frequency problem, usually corrected with Laplace/add-one smoothing). The Minimum Description Length (MDL) principle instead selects the hypothesis that minimizes the combined cost of encoding the model plus encoding the data given the model, naturally penalizing overly complex feature sets and reducing overfitting — effectively an information-theoretic form of Occam's Razor applied to feature selection.

Application of Knowledge

- MLE-based feature selection tends to include many high-frequency but low-signal words, increasing model complexity without proportionate accuracy gain.
- MDL-based selection keeps only features that meaningfully reduce description length, yielding a more compact, generalizable spam filter that is less prone to overfitting the training corpus.
- The trade-off is bias-variance: MLE/large hypothesis spaces reduce bias but increase variance on unseen mail; MDL/smaller hypothesis spaces increase bias slightly but improve generalization.

Completeness of Response

Sample size directly affects accuracy: with few labelled emails, a smaller hypothesis space (as encouraged by MDL) generalizes better, consistent with sample complexity theory (more complex hypothesis spaces require proportionally more training examples to bound generalization error, per PAC-learning bounds). As the training corpus grows, richer hypothesis spaces (larger vocabularies, n-gram features) become viable without overfitting.

Clarity & Presentation

Overall, a Naive Bayes filter with MDL-guided feature selection and Laplace smoothing offers the best practical balance of accuracy, computational efficiency, and robustness to the evolving vocabulary used by spammers, while the Bayes Optimal Classifier remains a useful theoretical upper bound against which real system performance can be benchmarked.

Q3. Customer Segmentation using the EM Algorithm

Conceptual Understanding

The Expectation-Maximization (EM) algorithm is well suited to customer segmentation because purchasing data often has hidden (latent) structure — the true segment membership of each customer is unobserved. EM models the data as a mixture (typically a Gaussian Mixture Model) of k underlying customer segments, each with its own distribution over features such as purchase frequency, average order value, and category preference.

Accuracy of Answer

The E-step computes the posterior probability (responsibility) that each customer belongs to each segment, given the current parameter estimates. The M-step then re-estimates each segment's parameters (mean, covariance, mixing weight) by maximizing the expected log-likelihood using those responsibilities. These two steps alternate until the log-likelihood converges, i.e., successive improvements fall below a threshold.

Application of Knowledge

- Convergence behavior: EM guarantees monotonic non-decrease of the likelihood each iteration, but it converges to a local, not necessarily global, optimum, so results depend on the quality of the model fit.
- Initialization challenges: poor initial cluster centers (e.g., via random initialization) can lead to slow convergence or poor local optima; k-means++ style initialization or multiple random restarts mitigate this.
- Probability learning here means the model continuously refines its estimate of each customer's segment membership probability as more transaction data becomes available, rather than making a one-time hard assignment.

Completeness of Response

The soft-clustering output (probability of belonging to each segment) can be used for targeted marketing by ranking customers by their probability of belonging to a high-value segment and tailoring offers accordingly, while customers with high uncertainty (near-equal probabilities across segments) can be flagged for further behavioral data collection before targeting.

Clarity & Presentation

Limitations to note: EM assumes a specific parametric form (e.g., Gaussian) for each segment which may not match real purchasing-behavior distributions; it requires the number of segments k to be chosen in advance (via criteria such as BIC/AIC); and it can be sensitive to outliers and missing/incomplete transaction records, all of which should be validated against business knowledge before deploying the segmentation for marketing.

Q4. Concept Learning for Medical Image Classification

Conceptual Understanding

Concept learning frames the problem as searching a hypothesis space H for a hypothesis h that best approximates the true target concept c: normal vs. abnormal image. Each hypothesis is expressed as a conjunction of constraints over extracted image features (e.g., texture, density, shape irregularity). The Candidate-Elimination algorithm maintains a version space bounded by the most General boundary (G) and most Specific boundary (S) consistent with all training examples seen so far.

Accuracy of Answer

As each labelled image arrives, positive (abnormal) examples generalize the S boundary just enough to cover the new example, while negative (normal) examples specialize the G boundary to exclude it, progressively narrowing the version space toward hypotheses consistent with all observed data.

Application of Knowledge

- Finite hypothesis spaces (e.g., a fixed set of discretized feature-threshold rules) allow exact bounds on the number of examples needed to converge, since sample complexity scales with log|H|.
- Infinite hypothesis spaces (e.g., continuous-valued feature thresholds) require complexity measures such as VC-dimension to bound generalization error instead of simply counting hypotheses.

- The Mistake Bound Model evaluates how many prediction errors an online learner (such as a Halving or Weighted-Majority-style algorithm) makes in the worst case before converging to the correct concept, which is a useful proxy for how quickly a diagnostic-support tool becomes reliable as it is used clinically.

Completeness of Response

Sample complexity results indicate that more expressive (larger) hypothesis spaces need proportionally more labelled medical images to guarantee low error with high confidence (per PAC bounds), which is a genuine constraint in medical imaging where expert-labelled data is scarce and expensive to obtain.

Clarity & Presentation

Generalization to unseen images is best when the version space converges to a single, tightly bounded hypothesis; a version space that remains wide (S and G far apart) signals that more diverse training examples are needed. In practice, Candidate-Elimination is noise-intolerant, so it is best combined with robust feature engineering and, for noisy real-world scans, supplemented by probabilistic classifiers rather than used alone.

Q5. Bayesian Investment Decision System under Market Uncertainty

Conceptual Understanding

Bayesian inference treats market parameters (expected returns, volatility, correlations) as random variables with a prior distribution reflecting existing financial theory and historical data. As new market data (prices, earnings reports, macroeconomic indicators) arrives, Bayes' Theorem updates these into a posterior distribution: $P(\text{Parameters}|\text{New Data}) \propto P(\text{New Data}|\text{Parameters}) \cdot P(\text{Parameters})$, allowing the model to continuously refine its beliefs rather than relying on a single fixed estimate.

Accuracy of Answer

Where the posterior distribution cannot be computed in closed form (common with complex, multi-asset models), the Gibbs sampling algorithm is used: it iteratively samples each parameter from its conditional distribution given the current values of all other parameters, and after sufficient iterations the samples approximate draws from the true joint posterior distribution.

Application of Knowledge

- Prior knowledge (e.g., historical asset-class returns, analyst forecasts) anchors predictions when data is scarce, and its influence shrinks automatically as more market evidence accumulates.
- Risk is evaluated using the spread (variance/credible intervals) of the posterior distribution over expected returns, giving a probabilistic risk measure (e.g., probability of loss exceeding a threshold) rather than a single point estimate.
- As new data arrives, the model performs online/sequential Bayesian updating, allowing investment decisions to adapt to regime changes (e.g., shifts from bull to bear market conditions) instead of assuming a static environment.

Completeness of Response

Computational challenges include the high dimensionality of multi-asset portfolios, which makes Gibbs sampling slow to converge (poor mixing) when parameters are highly correlated, and the difficulty of choosing uninformative-yet-realistic priors for new or thinly-traded assets.

Clarity & Presentation

Optimization strategies to address these challenges include using conjugate priors where possible to obtain closed-form updates, parallelizing independent Markov chains to speed up sampling, and using variational inference as a faster approximate alternative to Gibbs sampling when near-real-time investment decisions are required. Together, these techniques let the Bayesian system balance predictive accuracy, computational cost, and responsiveness to changing market conditions.